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Mechanism of anaerobic digestion of cattle manure enhanced by oxygen and hydrogen nano-bubble water

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ABSTRACT

This study investigates the efficacy and mechanisms of hydrogen/oxygen nano-bubble water (H_2/O_2-NBW) in enhancing anaerobic digestion (AD) performance. Batch experiments demonstrated that H_2-NBW and O_2-NBW significantly increased methane yields by 4.21 % to 16.98 % and 3.37 % to 13.43 %, respectively, compared to the control group ($p < 0.05$). Partial least squares path modeling and random forest algorithms identified 100 % H_2-NBW and 80 %O_2-NBW as optimal dosages. Under these optimal dosage conditions, semi-continuous tests revealed that methane production rates increased by 4.9-fold for H_2-NBW and 4.1-fold for O_2-NBW at an organic loading rate of 4 g volatile solids (VS)/(L·d) compared to the control group, while maintaining a stable output of 157.01–163.66 mL/gVS·d at 5 g VS/(L·d) when the control failed. Mechanistic investigations indicated that O_2-NBW generates hydroxyl radicals that reduce cellulose crystallinity by 39.81 % and upregulate hydrolytic enzymes such as xylanase, cellulase, protease, and α-glucosidase, targeting the regulation of hydrolytic and acidogenic bacteria (*Sedimentibacter*). In contrast, H_2-NBW activates hydrogenotrophic methanogenesis through the stimulation of dehydrogenase, acetokinase, and coenzyme F_{420}, along with the enrichment of *DMER64* and *Methanobacterium*. Microbial co-occurrence network analysis indicated that the addition of NBW significantly altered microbial interaction patterns. Functional predictions confirmed that H_2/O_2-NBW specifically strengthened the putative metabolic pathways of hydrogenotrophic methanogenesis and hydrolysis-acidogenesis. The synergistic mechanism of physical deconstruction and biological regulation of nanobubble water revealed in this study has opened a new pathway for the development of technologies aimed at enhancing AD performance.

1. Introduction

In 2023, China's rapidly expanding livestock sector produced 97.48 million tons of meat, 35.63 million tons of poultry eggs, and 42.81 million tons of dairy products [1]. This intensive farming practice generates approximately 3.8 billion tons of manure annually; however, less than 60 % of this waste undergoes proper treatment [2]. Consequently, over 1.5 billion tons of agricultural waste are released into the environment untreated each year. Such crude disposal methods not only squander nitrogen, phosphorus, and potassium resources but also precipitate cascading environmental risks, including heavy metal accumulation, eutrophication of water bodies, and greenhouse gas emissions, which have become significant bottlenecks hindering the green transition in agriculture. Under China's "dual carbon" strategy, which aims to achieve carbon peak by 2030 and carbon neutrality by 2060, anaerobic digestion (AD) technology—known for its synergistic benefits of organic matter degradation, energy recovery, and carbon emission reduction—has been recognized as a crucial solution for addressing the challenges of agricultural waste management [3]. However, the unique three-dimensional heterogeneous structure of lignocellulose, particularly the encapsulation effect of lignin on cellulose and hemicellulose, results in inefficient hydrolysis, with actual methane yields often falling below 60 % of the theoretical maximum [4]. Existing process enhancement techniques, such as co-digestion [5], bioaugmentation [6], and the application of chemical additives [7], can improve degradation efficiency to a certain extent. However, these methods are often subject to risks, including seasonal instability in substrate supply, high costs associated with microbial agents and chemical reagents, and potential secondary pollution resulting from chemical agents. These factors significantly limit their engineering applications. Therefore, the development of energy-efficient and environmentally friendly enhancement technologies has become crucial for optimizing AD system

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performance.

In recent years, micro-aeration technology has been extensively reported to enhance lignocellulosic hydrolysis [8], while the introduction of hydrogen into anaerobic digesters is widely recognized for biogas upgrading [9]. However, the direct introduction of gases such as O_2 and H_2 in gaseous form into AD systems presents challenges, including low apparent solubility and short retention time. To mitigate spikes in dissolved oxygen or sudden increases in hydrogen partial pressure caused by rapid gas escape—which can inhibit the activity of strict anaerobic methanogens and syntrophic microorganisms—precise dosing of these gases is crucial. This requirement undoubtedly complicates engineering operations. In contrast, incorporating gases into the liquid phase through specialized methods to create nano-bubble water (NBW) has emerged as a promising alternative [10]. NBW is an aqueous system characterized by a high density of stable submicron-sized bubbles (typically <1 μm), featuring a high specific surface area, low buoyancy, and prolonged residence periods, which together facilitate highly efficient gas-liquid synergistic interfaces [11,12]. As a liquid-phase additive, NBW is easy to handle and significantly enhances the apparent solubility and retention time of poorly soluble gases such as H_2 and O_2. Thus, NBW can serve as an efficient gas delivery and sustained-release platform, effectively targeting various bottlenecks in AD. For instance, oxygen nano-bubble water (O_2-NBW) can establish a stable micro-oxygen environment, stimulate facultative hydrolytic bacteria, and promote the enzymatic breakdown of recalcitrant lignocellulose, thereby overcoming the rate-limiting step of hydrolysis [10]. Meanwhile, hydrogen nano-bubble water (H_2-NBW) is anticipated to provide a continuous and stable electron donor (H_2) for hydrogenotrophic methanogens, thereby accelerating the CO_2 reduction methanogenesis process and directly increasing methane yield [13]. Additionally, the collapse of nano-bubbles may generate localized high energy and free radicals (e.g., ·OH), potentially disrupting the lignin-carbohydrate complex and further enhancing substrate bioavailability [10]. Therefore, the enhancing effect of NBW arises from the intrinsic synergy between its role as a functionalized gas carrier and its physico-chemical effects, which extend far beyond simple gas-water mixing, enabling precise regulation of microbial metabolic pathways and reaction processes in AD systems [14,15].

From an engineering application perspective, the integration of NBW technology into existing AD systems is highly feasible. A key advantage of this technology lies in its ability to replace conventional dilution water (e.g., tap water, recycled water) used for adjusting feedstock solid content in existing processes with functionalized NBW, without altering the system's original hydraulic loading or feedstock solid concentration, and without increasing overall water consumption. More importantly, nano-bubble generators have evolved into highly modular and commercially mature equipment [8], which can be seamlessly integrated into existing dilution water pipelines for on-demand production and automated dosing. This significantly enhances the engineering practicality and marketability of this technology. However, despite numerous studies documenting the beneficial effects of NBW addition on AD performance [10,16,17], most existing research is confined to batch experiments. There is a notable lack of systematic understanding regarding the long-term effects of NBW addition on the mechanisms governing the entire four-stage AD process. This is particularly true concerning its regulation of functional microbial community structure, dynamics of key enzyme activities, and metabolic pathways. This gap significantly hinders further optimization of the process and the large-scale application of NBW enhancement technology.

This study systematically investigates the enhancement mechanism of H_2/O_2-NBW on the AD process of cattle manure, addressing the identified scientific bottlenecks. Batch experiments with varying dosages were designed to determine the optimal dosing ratios for H_2-NBW and O_2-NBW. Subsequently, long-term continuous experiments were conducted to elucidate the regulatory patterns of NBW on enzyme activities related to hydrolysis and methanogenesis. Finally, an integrated microbial analysis was performed to clarify the reconfiguration mechanisms of the functional microbial community structure and metabolic pathways. By coupling enzymatic activity dynamics with microbial succession analysis, this study elucidates the long-term enhancement mechanisms, aiming to provide theoretical foundations and engineering-oriented optimization strategies for the valorization of agricultural waste.

2. Materials and methods

2.1. Inoculum and substrate

The inoculum sludge utilized in this experiment was sourced from the anaerobic digester of the Tangjiatuo Wastewater Treatment Plant in Chongqing, China. After retrieval, the sludge was allowed to settle for three days to eliminate large inorganic particles, followed by centrifugation to concentrate its total solids (TS) from 3 % to 8 %. The concentrated sludge was subsequently placed in a continuous stirred tank reactor with an effective volume of 15 L for acclimatization. Initially, no feeding was conducted to deplete the background organic matter present in the inoculum. Once the daily gas production fell below 1 L, cattle manure was introduced as the substrate. Acclimatization occurred at an organic loading rate (OLR) of 3 g VS/L·d and a hydraulic retention time (HRT) of 30 days until the daily gas production fluctuated by no more than 10 % over a consecutive week, indicating the successful completion of sludge acclimatization [18]. Following this, continuous feeding was maintained under the original acclimatization conditions to sustain microbial activity for subsequent experiments. The cattle manure was collected from a small dairy farm in Chongqing and stored in a refrigerator at 4 °C for future use. The physicochemical properties of the acclimated inoculum sludge and substrate are presented in Table S1.

2.2. Nano-bubble water

In this study, O_2-NBW and H_2-NBW were produced using a nano-bubble water generator (Model RWP180, Ruke Environmental Protection Equipment Co., Ltd., Jiangsu, China) through the gas-water circulation method [19]. Oxygen gas (purity ≥99.995 %) was obtained from Tianjin Anxing Industrial Gas Sales Co., Ltd. (China), while hydrogen gas (purity ≥99.995 %) was generated using a hydrogen generator. A total of 7.5 L of deionized water was circulated at a flow rate of 2.5 L/min, with the gas intake rate and pressure of the nano-bubble generator maintained at 0.07 L/min and 0.24 MPa, respectively. Continuous infusion of oxygen or hydrogen was conducted for 20 min to produce O_2-NBW and H_2-NBW [16,20]. The resulting NBW exhibited the following properties: average bubble sizes of 183.54 ± 24.32 nm and 214.39 ± 35.21 nm, bubble concentrations of 2.30×10^7 bubbles/mL and 2.40×10^7 bubbles/mL, and zeta potentials of −31.65 ± 2.98 mV and − 23.90 ± 1.32 mV for O_2-NBW and H_2-NBW, respectively. These key properties align with those reported in the literature [16,21,22], thereby confirming the robustness of our methodology and the validity of comparing our results with existing research. Additionally, the dissolved oxygen content of O_2-NBW and the dissolved hydrogen concentration of H_2-NBW were measured at 37.30 mg/L and 1360 ppb, respectively.

2.3. Batch experimental setup

Batch experiments were conducted using 500 mL serum bottles with a working volume of 400 mL. All reactors were established by adding cattle manure as the substrate and pre-acclimated inoculum sludge, which had been pre-incubated until gas production ceased, at a VS ratio (S/I) of 1.5. The TS concentration of the reactor contents was adjusted from 12.06 % to 8 % by adding dilution water, with a total of 111.30 mL added to each reactor. The control group utilized ultrapure water as the dilution medium, while the ten experimental groups received 20 %, 40 %, 60 %, 80 %, and 100 % (volume ratio in dilution water) of either H_2-

NBW or O_2-NBW, with three replicates for each experimental group. This experimental design aimed to investigate whether the use of functionalized NBW as dilution water can enhance the AD process compared to conventional dilution water. After a 5-min nitrogen purging, the serum bottles were sealed and incubated in a thermostatic water bath shaker at 37 °C with an agitation speed of 100 rpm. Samples from the reactors were collected on the 4th, 8th, 12th, 16th, 20th, and 25th days of the experiment to determine physicochemical parameters. The samples collected on the 25th day were additionally utilized to assess enzymatic activity. Furthermore, gas samples were collected daily using a syringe for gas composition analysis.

2.4. Gas sparging verification experiment

Following the identification of optimal dosage ratios for the two types of nanobubble water (100 %H_2-NBW and 80 %O_2-NBW) in the preceding experiments, a supplementary sparging control experiment was conducted. This experiment aimed to ascertain whether the enhanced reactor performance is attributable to the gases carried by the NBW or to the inherent synergy between its role as a functionalized gas carrier and its physicochemical effects. The reactor setup, S/I ratio, TS content of the mixture, and other experimental conditions were consistent with those described in Section 2.3. The configurations of the control group, 100 %H_2-NBW, and 80 %O_2-NBW groups also mirrored those in Section 2.3. Additionally, four sparging groups were established, consisting of two groups for H_2 and two groups for O_2. Among these, two groups (H_2 sparging at 22.81 mL and O_2 sparging at 18.25 mL) received aeration volumes equivalent to the total gas input during the generation of NBW, as calculated based on the preparation procedure detailed in Section 2.2. The remaining two groups (H_2 sparging at 1.86 mL and O_2 sparging at 2.55 mL) received aeration volumes that corresponded to the amount of gas entering the AD reactor via the NBW, which was back-calculated according to the measured dissolved gas concentrations in NBW. The detailed experimental design is illustrated in Fig. S1a. The aeration rate for the sparging groups was aligned with the gas injection rate of the nanobubble generator. Sparging was executed using a peristaltic pump (Rong Bai, BT100-2J), and all reaction conditions and gas sampling procedures adhered to those outlined in Section 2.3.

2.5. Semi-continuous reactor operation

To investigate the long-term effects of dilution water addition on reactor performance, ultrapure water, 100 %H_2-NBW, and 80 %O_2-NBW were selected as dilution waters for semi-continuous verification experiments. A semi-continuous experiment was conducted using three acrylic continuous stirred-tank reactors, each with a working volume of 15 L. Feeding and discharging occurred once daily. The acclimation of the inoculum and the preparation of the substrate are detailed in Section 2.1. The reactor temperature was maintained at 37 ± 1 °C using a thermostatic control system. A top-mounted agitator operated at 100 rpm for 4-h intervals, followed by 2-h pauses.

During the start-up phase (Phase I), the OLR was set at 3 g VS/(L·d), with an HRT of 30 days. The total mass of the feed was calculated based on the reactor's effective volume, OLR, and substrate characteristics. Subsequently, the digesters were fed after diluting the substrate with water to achieve a TS content of 10 %. Gas production from each reactor was measured daily to monitor operational status, while liquid samples were collected every two days to assess various physicochemical parameters. The transition between reactor phases and the increase in OLR adhered to performance-based criteria reported in the literature [23,24], where a pseudo-steady state was considered achieved if the daily biogas production fluctuation remained within 10 % for seven consecutive days. Upon meeting this criterion and ensuring that the reactor had operated for more than one HRT, the OLR was increased to 4 g VS/(L·d) (Phase II). After maintaining stable operation for no less than one HRT at an OLR of 4 g VS/(L·d), the OLR was further elevated to 5 g VS/(L·d) (Phase III). Throughout each phase, dilution water was introduced into the reactor alongside the substrate during the daily feeding operations for the entire experimental period. Microbial samples were collected at the conclusion of Phase I, Phase II, and Phase III, and stored at −80 °C in an ultra-low temperature refrigerator for subsequent microbial community analysis.

2.6. Analytical methods

The methods employed to monitor methane content and the concentrations of VFAs — including acetate, propionate, butyrate, isobutyrate, valerate, and isovalerate — along with pH, total alkalinity (ALK), partial alkalinity (PA), TS, VS, chemical oxygen demand (COD), ORP, ZETA potential, total ammonia nitrogen (TAN) concentration, soluble polysaccharides (PS), and soluble proteins (PN) were based on procedures described in previous studies [25,26]. The cellulose content in cattle manure was determined using the Van Soest method. The activities of xylanase, acetate kinase, protease, coenzyme F_{420}, α-glucosidase, cellulase, and dehydrogenase were assessed according to methodologies outlined in prior researches [27,28]. The crystallinity index of cellulose in the samples, both before and after the AD experiment, was evaluated based on the technique described by Wang et al. [29] The diluted water from the three experimental groups (i.e., the control group, 80 %O_2-NBW group, and 100 %H_2-NBW group) was analyzed for hydroxyl radicals present in NBW using an EMXmicro-9.5/12/P/L type instrument for electron paramagnetic resonance spectroscopy prior to being added to the reactor [30]. The concentrations of dissolved hydrogen in H_2-NBW and dissolved oxygen in O_2-NBW were measured directly using a portable dissolved hydrogen meter (Kedida, CT-8023 L) and a portable dissolved oxygen meter (Leici, JPBJ-608), respectively.

2.7. Kinetic analysis

The modified Gompertz equation (Eq. 1), as proposed by Ye et al. [26], is widely utilized for complex substrates to simultaneously analyze the lag phase and the biogas production rate. This equation was employed to assess methane production kinetics.

$$M_{(t)} = M_{max}\exp\left\{-\exp\left[\frac{R_{max}\times e}{M_{max}}\times(\lambda - t)+1\right]\right\} \quad (1)$$

where $M_{(t)}$ represents the methane yields at time t, mL/g VS; M_{max} represents the methane yield potential, mL/g VS; t represents the digestion time, d; R_{max} represents the maximum methane production rate, mL/g VS; λ represents the lag phase, d; e represents the natural logarithm, exp (1) = 2.7183.

2.8. Microbial community structure analysis

Genomic DNA extraction was performed using the FastDNA®SPIN Kit for Soil (MP Biomedicals). The variable regions of the 16S rRNA genes were amplified via PCR employing modified universal primers 338FmodF (GTGYCAGCMGCCGCGGTAA) and 806RmodR (GGACTACNVGGGTWTCTAAT). The resulting amplicons were sequenced on an Illumina MiSeq platform (Majorbio). Bioinformatics processing encompassed raw data quality control, contig assembly, reference-based taxonomy annotation, α-diversity assessment, and community structure analysis, in accordance with established bioinformatics protocols from previous studies [25,31].

2.9. Statistical analyses

A one-way analysis of variance (ANOVA) was conducted to test the differences in physicochemical indices among the groups. Descriptive

statistics (means ± standard deviation) and Spearman correlations were generated using SPSS. The following analyses were performed using various R packages in RStudio. The partial least squares path model (PLS-PM, using the 'plspm' package) quantified both linear direct and indirect effects of physicochemical properties, organics, VFAs and enzyme activity on methane production. Random forest analysis (RFA, utilizing the 'randomForest' and 'rfPermute' packages) captured non-linear relationships, with variable importance assessed via percentage increase in mean squared error (%IncMSE). Co-occurrence networks were visualized using the 'igraph' and 'psych' packages to illustrate interactions among all taxa. Mantel tests, performed with the 'Hy4m/linkET' package, correlated methane yield (measured by Euclidean distance) with Bray-Curtis dissimilarity of differential microorganisms.

3. Results and discussion

3.1. Effect of nano-bubble water types and dose on reactor performance

3.1.1. Biogas production performance and kinetics

As illustrated in Fig. 1, the incorporation of H_2-NBW and O_2-NBW demonstrates differentiated enhancing effects on biogas production performance during the AD of cattle manure. In the H_2-NBW system, the cumulative methane yield (CMY) of each experimental group showed a significant positive correlation with the addition ratio of H_2-NBW in dilution water (correlation coefficient $r = 0.756$, $p < 0.001$, Fig. 1c). At a 100 % dosage, the peak daily methane production increased by 9.48 % compared to the control group (Fig. 1a), and the CMY reached 286.13 mL/g VS, representing a 16.98 % increase over the control group (Fig. 1b). This observation aligns with previous studies indicating that H_2-NBW can gradually release H_2 [32], thereby providing a substrate for methanogenesis. Notably, the dissolved hydrogen concentration in H_2-NBW was measured at 1360 ppb. Even with the 100 %H_2-NBW dosage, the total amount of hydrogen introduced into the reaction system alongside 111.30 mL of dilution water, was only 1.70 mL (under standard conditions). According to the stoichiometry of hydrogenotrophic methanogenesis ($4H_2 + CO_2 \rightarrow CH_4 + 2H_2O$), this quantity of hydrogen could theoretically yield only 0.425 mL of CH_4. This amount is significantly lower than the methane production increments (474.74 mL) observed in the 100 %H_2-NBW group compared to the control group, indicating that the contribution rate of H_2 as a substrate to the gas production increment is less than 0.1 %. This clearly suggests that the enhancement in methane production primarily arises from the unique properties of the nanobubble carrier itself, rather than its role in introducing a reactive substrate. The results from the sparging experiments further substantiate this inference (Fig. S1b). The CMY in the high and low-dosage groups increased by 4.61 % and 2.69 %, respectively, compared to the control group. The gas input volume in the low-dosage group was consistent with that of the 100 %H_2-NBW group; however, the differences in gas delivery methods resulted in significant variations in gas production increments. This indicates that nanobubbles do not merely serve as gas 'transport vehicles'; rather, they likely enhance the activity of methanogenic archaea more effectively through mechanisms such as stabilizing micro-interfaces, improving mass transfer, or directly interacting with microorganisms. Although the high-dosage group received a greater volume of gas input, it exhibited a pronounced lag period in gas production during the initial reaction phase. This observation demonstrates that H_2 input via sparging is highly susceptible to inducing a high hydrogen partial pressure, which inhibits gas production, thereby underscoring the advantages of nanobubble water-mediated gas delivery. Notably, even if the gas volumes from the high and low-dosage groups were entirely utilized for methane production, the theoretically calculated methane increments would only be 5.70 mL and 0.465 mL, respectively, still representing less than 5 % of the total gas production increase in each group. This illustrates that whether H_2 is introduced via nanobubble water or sparging, the contribution rate of H_2 as a substrate to the gas production increment remains extremely low.

In contrast, the O_2-NBW group exhibited a non-monotonic response

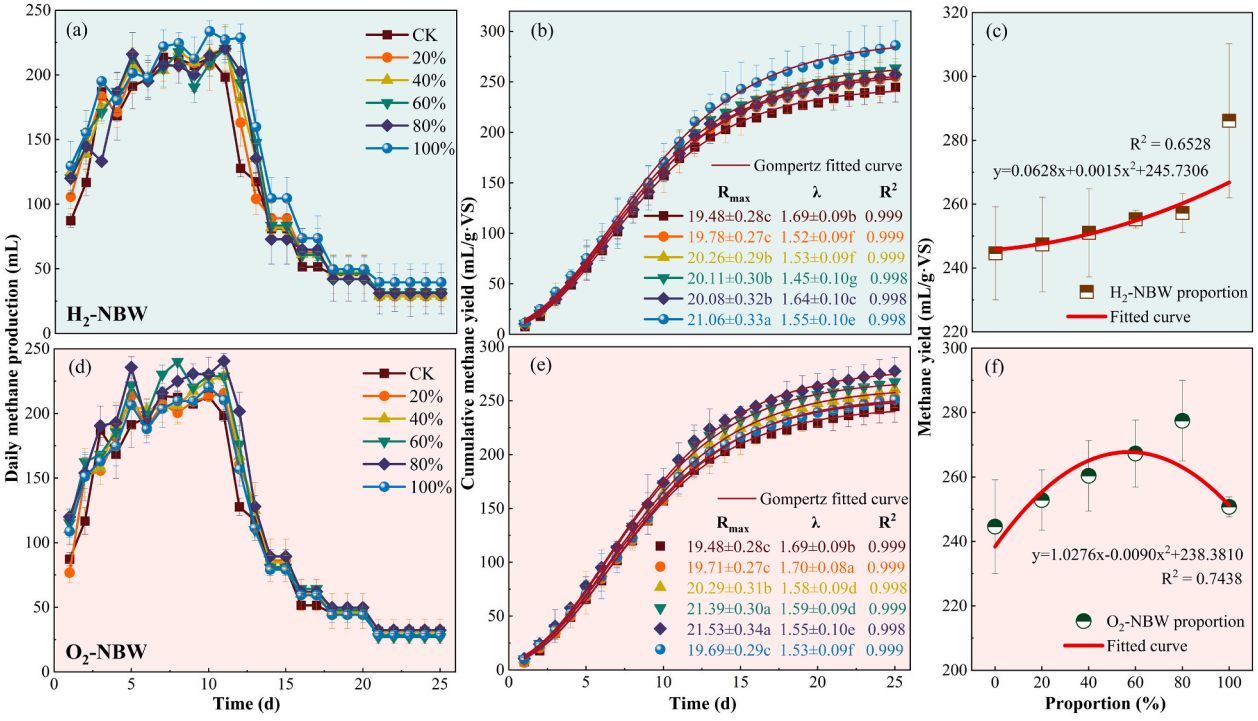


Fig. 1. Methane production characteristics in (a-c) H_2-NBW and (d-f) O_2-NBW treatment groups: (a, d) daily methane production, (b, e) CMY alongside kinetic parameters, and (c, f) the response of CMY to varying proportions of NBW. In subgraphs (b) and (e): scatter plots represent the measured gas production data, with red solid lines denoting the model-fitted results. Statistically significant differences are indicated by different superscript letters adjacent to the kinetic parameters ($p < 0.05$, ANOVA).

characterized by a 'low ratio promotion and high ratio inhibition' (Fig. 1f). Specifically, the 80 %O_2-NBW group achieved the highest peak daily methane production of 240.49 mL/d, representing a 21.30 % increase over the control group (Fig. 1d), while the 100 %O_2-NBW group only reached 210.29 mL/d. This phenomenon may be attributed to the excessive proportion of O_2-NBW, which elevates the ORP in the reactor (measured ORP: −163.09 ± 3.16 mV [33]), deviating from the optimal ORP range for methanogens (below −300 mV [34]). At the end of the experiment, the CMY of the 80 %O_2-NBW group was the highest, reaching 277.45 mL/g VS, which was 13.43 % and 10.65 % higher than that of the control group (244.59 mL/g VS) and the 100 %H_2-NBW group (250.74 mL/g VS), respectively (Fig. 1e). These findings indicate that the appropriate addition of O_2-NBW can significantly enhance methane yield during the AD of cattle manure. Furthermore, the results from the oxygen sparging experiments further validate the substantial advantage of using NBW as a gas carrier over direct sparging. As illustrated in Fig. S1b, when a small volume of O_2 (2.55 mL, equivalent to the amount introduced via the 80 %O_2-NBW) was supplied to the reactor through sparging, the O_2 sparging group exhibited no stimulatory effect on methanogenesis, with the CMY decreasing by 0.43 % compared to the control group. In contrast, the CMY in the O_2 sparging group was only 8.91 % of that in the control group when the amount of O_2 was equal to the total gas volume used in the NBW preparation process. This indicates that, for both hydrogen and oxygen, sparging an equivalent volume of gas resulted in a lesser enhancement of methane production compared to the NBW groups. Moreover, when introduced at volumes matching the total gas used in NBW preparation, sparging caused varying degrees of inhibition in either methane production or methane production kinetics. In conclusion, utilizing NBW as a gas carrier and leveraging its gas-liquid synergistic interface leads to improved reactor performance compared to sparging, with the primary contribution arising from the holistic effect of NBW rather than the introduced gases serving as direct substrates for methane production.

The modified Gompertz model was employed to fit the biogas production kinetics across various AD systems, achieving a coefficient of determination (R^2) of ≥0.998. As illustrated in Fig. 1b, the R_{max} of the 100 %H_2-NBW group was measured at 21.06 mL/g VS, reflecting a significant increase of 8.11 % compared to the control, which recorded 19.48 mL/g VS ($p < 0.05$). Notably, the λ exhibited a significant decreasing trend from 1.69 to 1.55 days ($p < 0.05$) with the increasing proportion of H_2-NBW, suggesting that the introduction of H_2 accelerated the microbial community's adaptation to the system [35]. The kinetic response of the O_2-NBW group exhibited a nonlinear characteristic, as shown in Fig. 1e. The R_{max} for the 80 %O_2-NBW group significantly increased by 10.52 %, reaching 21.53 mL/g VS compared to the control group (19.48 mL/g VS, $p < 0.05$). However, when the proportion of O_2-NBW increased to 100 %, R_{max} decreased to 19.69 mL/g VS, showing no significant difference from the control ($p > 0.05$). This further indicates that a high proportion of O_2-NBW may diminish its potential to enhance methane production, potentially due to an excessively high redox potential. The results preliminarily indicate that the enhancing effect of NBW is closely associated with the type and dosage of the gas it carries; thus, the optimal proportions of H_2-NBW and O_2-NBW should be maintained at 100 % and 80 %, respectively, to achieve maximum positive stimulation.

3.1.2. Characteristics of substrate degradation and enzyme activity

Differences in substrate degradation characteristics reflect the differential regulation by NBW during the hydrolysis and acidogenesis phases. Fig. 2a and b illustrate the variation patterns of soluble PN and PS across the experimental groups, respectively. Soluble PN and PS are hydrolyzed from complex organic materials in the substrate during the initial phase of AD. A comparison of concentrations between experimental groups on days 4 and 8 revealed that the soluble PS and PN concentrations in the H_2-NBW and O_2-NBW groups were generally higher than those in the control group. Specifically, the 100 %H_2-NBW and 80 %O_2-NBW groups exhibited significant increases in soluble PS and PN concentrations of 9.66 %–9.93 % and 17.98 %–27.73 % on day 4 ($p < 0.05$), and 12.44 %–15.02 % and 11.28 %–15.85 % on day 8 ($p < 0.05$), respectively. These results demonstrate that both types of NBW enhance substrate hydrolysis to varying degrees. Notably, the soluble PN and PS contents in all experimental groups exhibited an initial increase followed by a decrease over time. This trend significantly deviates from the typical monotonically decreasing pattern of soluble organic matter reported in previous studies [11]. This anomaly may arise from the transient solubilization of particulate organic matter, the release of microbial cell lysis products, or analytical interference caused by colloidal substances. The underlying mechanisms warrant further investigation. Fig. 2c illustrates the differences in removal efficiencies of cellulose, TS, and VS across the experimental groups. The results indicate that the removal efficiencies of cellulose, TS, and VS in the 100 %H_2-NBW and 80 %O_2-NBW groups were significantly elevated by 88.89 %, 21.37 %, 22.54 %, and 132.13 %, 25.91 %, 25.01 %, respectively, compared to the control group ($p < 0.05$). This finding reaffirms that the addition of H_2/O_2-NBW enhances substrate hydrolysis efficiency. Furthermore, the most substantial decrease in cellulose crystallinity was observed in the 80 %O_2-NBW group, which exhibited a reduction of 39.81 %. This reduction was 43.25 % and 14.86 % greater than that of the control group (27.79 %) and the 100 %H_2-NBW group (34.66 %), respectively (Fig. S2). This suggests that the introduction of O_2-NBW preferentially enhances the bioaccessibility of cellulose, thereby facilitating greater enzymatic hydrolysis and microbial utilization of cellulose [29]. Notably, the magnitude of the increase in these parameters used to assess the degree of substrate hydrolysis was significantly greater than that of the increase in CMY discussed in Section 3.1.1. This discrepancy may be attributed to the fact that, in addition to being utilized for acidogenesis, acetogenesis, and methanogenesis, hydrolyzed substrates are also employed for other purposes, such as the synthesis of extracellular polymeric substances. Strong evidence for this is that the soluble PN and PS in each experimental group were not completely metabolized at the final stage.

Electron paramagnetic resonance analysis revealed that the DMPO-OH adducts in O_2-NBW exhibited a characteristic quadruple-peak signal with a typical 1:2:2:1 ratio ($\alpha^N = 14.67$ G, $\alpha^H_\beta = 14.67$ G, peak intensity = 0.08 a.u.), confirming substantial ·OH generation during the collapse of O_2-NBW nano-bubbles (Fig. S3). The presence of hydroxyl radicals could enhance the accessibility of cellulose and reduce its crystallinity (39.81 %) by attacking the β-O-4 bond between lignin phenylpropane units [36], thereby disrupting the three-dimensional network structure of the complex. In contrast, no obvious quadruple peak signals were observed in the H_2-NBW group, likely due to scavenging by the direct reaction of hydrogen with free radicals ($H_2 + 2{\cdot}OH \rightarrow 2H_2O$) [37].

Fig. 2d illustrates the temporal variations in VFA concentrations across different reactors, resulting from the dynamic equilibrium between acid production and consumption in each reactor. In the 100 %H_2-NBW and 80 %O_2-NBW groups, acetate concentrations reached 6927.07 mg/L and 5469.31 mg/L on day 4, reflecting increases of 132.77 % and 83.79 %, respectively, compared to the control group (2975.88 mg/L). Notably, the VFA concentrations in both the 100 %H_2-NBW and 80 %O_2-NBW groups consistently exceeded those in the control group during the same timeframe, indicating that the addition of NBW significantly accelerates acidogenesis. It is important to recognize that acid production and methane production represent distinct metabolic stages, each dominated by different microbial communities. Therefore, the observed discrepancy in acid accumulation between the experimental and control groups, which contrasts with the methane production rates discussed in Section 3.1.1, is understandable. Furthermore, the amount of acid accumulated does not directly correlate with the amount of acid produced, thus the observed discrepancy in acid accumulation between the experimental and control groups, which is inconsistent with the differences in gas production, is also anticipated.

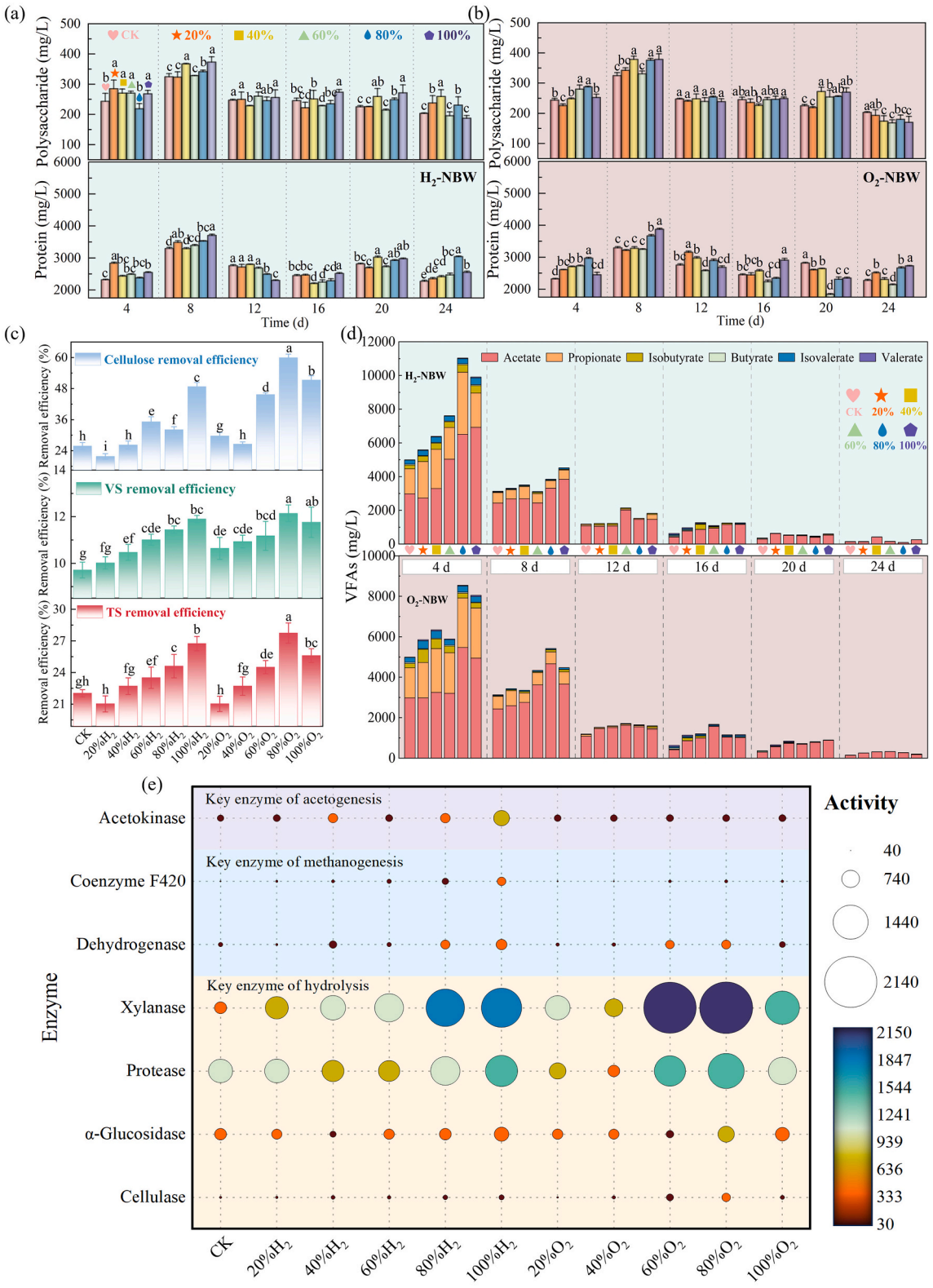


Fig. 2. Temporal variations in (a) the H_2-NBW and (b) O_2-NBW groups concerning dissolved protein and polysaccharide concentrations. (c) Cellulose, TS, and VS removal efficiencies across treatment groups (notation: 20 %H_2 denotes 20 %H_2-NBW group, among others). (d) VFA concentration dynamics. (e) Enzyme activities including xylanase, acetokinase, cellulase, and protease (measured in U/L), as well as α-glucosidase and dehydrogenase (measured in IU/L), and coenzyme F_{420} concentrations (measured in ng/L).

Fig. S4 illustrates the responses of pH, TAN, and PA to H_2-NBW and O_2-NBW across various concentration gradients. The dynamic evolution of system pH indicates that NBW enhances the acid-base buffering capacity of the AD process. Although the peak VFA concentrations in the H_2/O_2-NBW groups increased by 71.11 %–120.85 % compared to the control group, the pH remained consistently stable within the range of 7.4–7.7, effectively avoiding both acidic collapse (pH < 6.8) and alkaline inhibition (pH > 8.2). This stability may arise from the NH_4^+-NH_3 buffer system (TAN = 850–1050 mg/L), which maintains proton equilibrium [38]. The dynamics of PA further demonstrate the system's capacity to withstand shock loading. Throughout the experimental period, the PA concentrations in both the 100 %H_2-NBW and 80 %O_2-NBW groups exhibited an upward trend. On the 4th and 8th days, the VFAs/PA ratios for 100 %H_2-NBW and 80 %O_2-NBW groups were 2.09, 1.53 and 2.26, 1.66, respectively. At this time, daily methane production exhibited abnormal fluctuations and was lower than that of the control group (Fig. 1a and c). After day 8, biogas production stabilized and surpassed that of the control group. The VFAs/PA ratios for the two treatment groups ranged from 0.05 to 0.30, significantly below the acidification risk threshold of 0.5 [38]. This can be attributed to the effective role of the NBW in preventing system imbalance by facilitating the conversion of VFAs (Fig. 2d). Furthermore, the PA content of 80 % O_2-NBW surpassed that of 100 %H_2-NBW from day 8 onward, which can be attributed to the processes of free radical-mediated oxidation and neutralization. Hydroxyl radicals produced during the collapse of oxygen nano-bubbles may not only facilitate the degradation of lignocellulose but also oxidize a portion of VFAs, thereby reducing their accumulation. Simultaneously, the negatively charged surface of the nano-bubbles can adsorb and neutralize H^+, maintaining a weak alkalinity in the liquid environment and enhancing the PA content [39]. In contrast, hydrogen nano-bubbles are less effective in generating free radicals (Fig. S3), exhibit a weak oxidizing capacity (primarily hydrogen-reducing), and scavenge VFAs less efficiently than oxygen nano-bubbles.

The enzyme activity characteristics demonstrated a differential regulation of the rate-limiting steps in AD by H_2/O_2-NBW. As illustrated in Fig. 2e, at the end of the experiment, the activities of seven enzymes increased by 21.71 % to 635.84 % in the 100 %H_2-NBW group and by 8.79 % to 179.61 % in the 80 %O_2-NBW group compared to the control group. This finding suggests that the addition of NBW stimulates microbial metabolism. A key characteristic is that the addition of H_2-NBW specifically enhances enzyme activities related to hydrogen and methanogenic metabolism. The activities of dehydrogenase, coenzyme F_{420}, and acetokinase in the 100 %H_2-NBW group exceeded those in the 80 % O_2-NBW group by 17.41 %, 186.31 %, and 119.75 %, respectively. Among these, the increased activity of acetokinase accelerates the acetate activation step (acetate + adenosine triphosphate → acetyl phosphate + adenosine diphosphate), synergizing with the activity of phosphotransacetylase to facilitate the conversion of acetate to acetyl-CoA, thereby enhancing the flux of acetate metabolism [38]. This process is coupled with dehydrogenases to utilize VFAs and drive methane production. Coenzyme F_{420} provides essential metabolic support for hydrogenotrophic methanogens by mediating [H] transfer and activating the CO_2 reductase system [9]. In contrast, the addition of O_2-NBW specifically enhances the activity of hydrolytic enzymes through micro-oxygen pressure. The activities of cellulase, α-glucosidase, protease, and xylanase in the 80 %O_2-NBW group increased by 34.65 %, 12.91 %, 12.79 %, and 31.79 %, respectively, compared to the 100 %H_2-NBW group. This specific difference arises because H_2-NBW directly influences methanogenic metabolism by enhancing electron supply, while O_2-NBW improves hydrolysis efficiency during the initial phases of AD by increasing substrate availability.

3.1.3. Path modeling and non-linear interrelations

This study elucidates the differential regulatory mechanisms of the AD process influenced by H_2/O_2-NBW, utilizing PLS-PM in conjunction with RFA (Fig. 3). The model's goodness-of-fit (GoF > 0.72) and the explanatory power of the latent variables (R^2 > 0.7) confirm the reliability of the results. In the 100 %H_2-NBW group, model analysis revealed that the enhancement of key enzyme activities, such as coenzyme F_{420} and dehydrogenase, constitutes the primary pathway driving methane production. Specifically, the PLS-PM demonstrated a significant path coefficient of 0.470 (p < 0.05) from enzyme activity to methane production (Fig. 3a and c). Additionally, RFA model analysis indicated contributions of 6.11 % for coenzyme F_{420} (p < 0.001) and 6.04 % for dehydrogenase (p < 0.01) (Fig. 3e). These findings suggest that H_2-NBW enhances efficiency primarily by significantly increasing enzymatic activity in methanogenic terminal metabolism, particularly in hydrogenotrophic methanogenesis dependent on coenzyme F_{420}.

In stark contrast, the impact pathway of the 80 %O_2-NBW group was primarily characterized by an enhancement in the conversion of front-end substrate hydrolysis. The PLS-PM indicated that the physicochemical properties positively influenced the accumulation of VFAs, with a path coefficient of 0.891 (p < 0.001). This accumulation subsequently led to a significant increase in methane production, as evidenced by a path coefficient of 1.095 for the conversion from VFAs to methane (p < 0.05) (Fig. 3b). Furthermore, the RFA model substantiated the critical contributions of hydrolytic enzymes, revealing that cellulase contributed 7.68 %, xylanase 7.61 %, and protease 7.19 % to the process (p < 0.01). These findings suggest that O_2-NBW enhances methane production primarily by improving hydrolytic enzyme activity, thereby optimizing substrate deconstruction and acidogenesis efficiency. In summary, the model analyses reaffirm a fundamental divergence in the enhancement mechanisms between the two types of NBW: H_2-NBW primarily targets the direct optimization of terminal methanogenesis, whereas O_2-NBW focuses on augmenting front-end hydrolysis and acidogenesis efficiency. Based on the physicochemical and biochemical indicators, it is evident that 100 %H_2-NBW and 80 %O_2-NBW represent the optimal addition dosages.

3.2. Anaerobic fermentation performance in semi-continuous reactors

3.2.1. Methane production performance and process stability

Based on the results of the batch experiments presented in Section 3.1, the addition of 100 %H_2-NBW and 80 %O_2-NBW maximally promoted biogas production without introducing excessive hydrogen partial pressure or oxygen concentration that could adversely affect the microorganisms. Consequently, semi-continuous experiments were conducted using these optimal dosages. Fig. 4 illustrates the variations in biogas production performance and process stability across the reactors.

During Phase I, the reactors exhibited a gradual upward trend in biogas production over the first 10 days, likely due to the effects of pre-incubation on microbial activity prior to the initiation of the experiment. After this period, all reactors reached a stable operational state. Subsequently, the methane yield of the control group stabilized at 124.51 ± 11.35 mL/g VS·d. In the final 7 days of this phase, the methane yields of the H_2-NBW and O_2-NBW groups were recorded at 171.07 ± 6.68 mL/g VS·d and 177.81 ± 3.87 mL/g VS·d, respectively, representing an increase of 63.06 %–63.94 % compared to the control group. This clearly demonstrates the enhancement in biogas production performance attributable to the addition of NBW (Fig. 4a). Notably, while the methane yields of the two experimental groups were comparable, the H_2-NBW group exhibited a significantly higher methane content than the O_2-NBW group (Fig. 4c). Conversely, the daily methane production of the O_2-NBW group was significantly higher than that of the other two groups (Fig. 4b). This discrepancy can be attributed to the H_2 input in the H_2-NBW group, which facilitates hydrogenotrophic methanogenesis by reducing CO_2 to methane. In contrast, the O_2-NBW group releases trace amounts of O_2 or hydroxyl radicals through bubble collapse, which further degrade VFA precursors via hydrogen abstraction and decarboxylation reactions, ultimately mineralizing them into CO_2 [40].

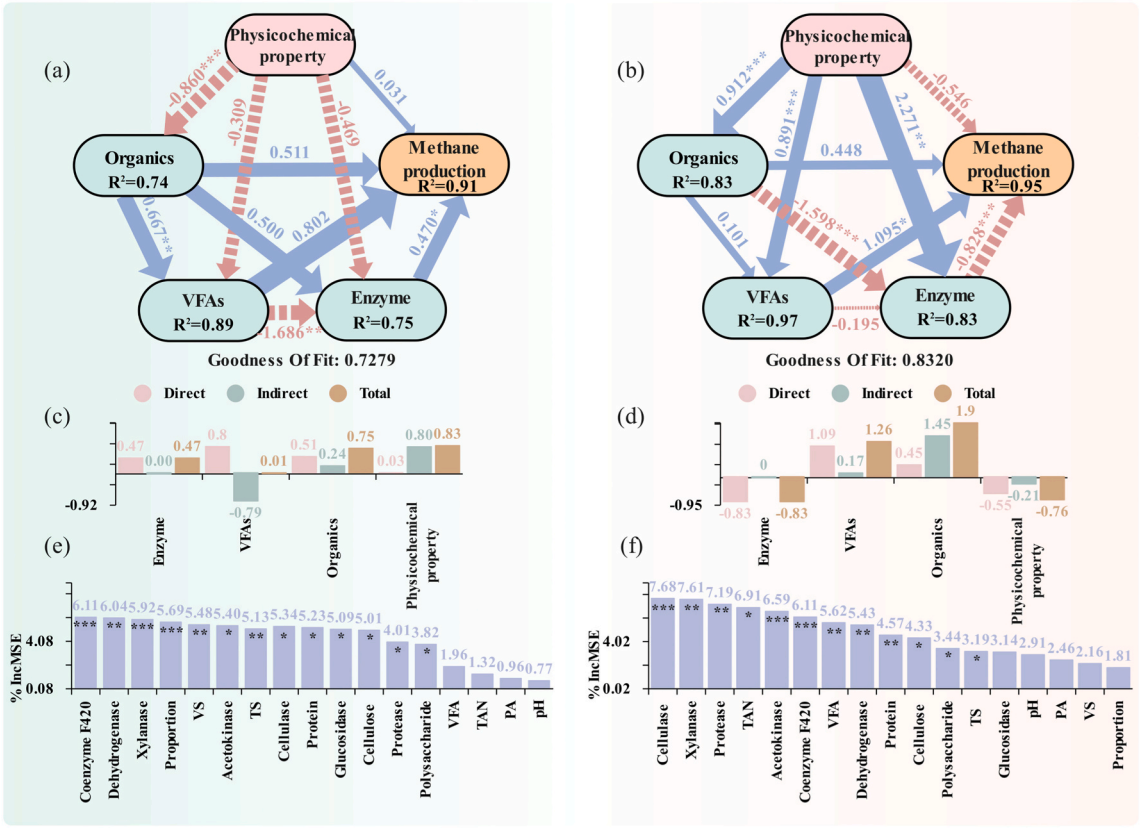


Fig. 3. PLS-PM analysis of treatment effects in (a) H_2-NBW and (b) O_2-NBW systems. Boxes represent observed (measured) variables or theoretical (latent) constructs: physicochemical properties (NBW proportion, pH, TAN, PA); Organics (VS, TS, cellulose, PN, PS); VFAs (acetate, propionate, butyrate, and valerate); and Enzymes (7 types shown in Section 2.5). Arrow values show standardized path coefficients (blue solid = positive, red dashed = negative effects; thickness = effect magnitude). GOF indicates model fit quality; R^2 values show construct explanatory power. (c-d) Standardized direct/indirect effects for H_2/O_2-NBW. (e-f) RFA of variable importance. Significance levels: *$p < 0.05$, ** $p < 0.01$, and ***$p \leq 0.001$.

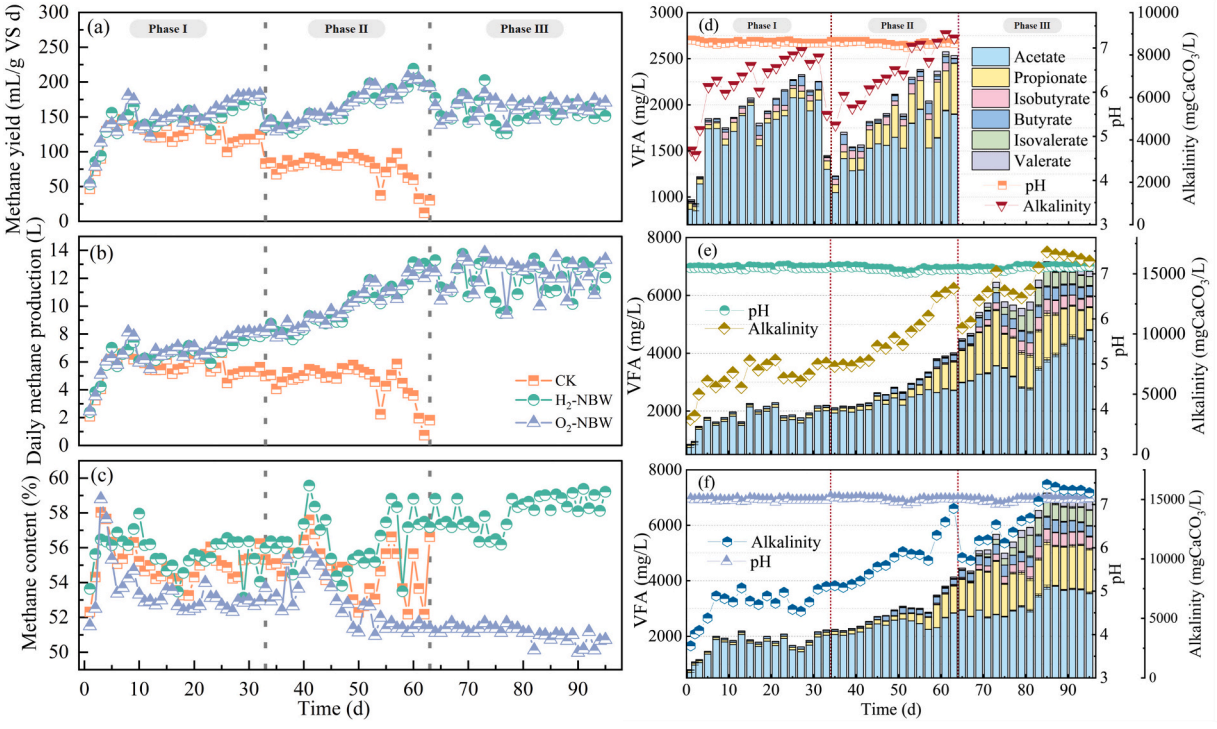


Fig. 4. Performance metrics across experimental groups: (a) methane yield, (b) daily methane production, (c) methane content, and (d-f) process parameters.

Despite the variations in biogas production characteristics, the VFA concentration in all reactors remained approximately 2000 mg/L during this phase, and the VFA/ALK ratio was significantly below 0.35 (Figs. 4d–4f), indicating stable operation across all reactors.

During Phase II, the enhancing effect of NBW on the biogas production performance of the reactors became increasingly pronounced. From days 33 to 56, while the methane yield of the control group remained relatively stable, it experienced a significant decrease of 34.47 % compared to its value in Phase I and was 46.20 %–46.69 % lower than that of the two experimental groups ($p < 0.01$). Beginning on day 57, a marked decline in biogas production was observed in the control group, whereas the methane yields of the two experimental groups remained stable at 195.49–195.58 mL/g VS·d during this period. Regarding physicochemical parameters, there was no significant difference in pH among the three reactors during this phase compared to Phase I. The total VFA concentration in the control group ranged from 930.50 to 2575.29 mg/L, maintaining a level similar to that in Phase I, with the VFA/ALK ratio fluctuating around 0.32, indicating that the acid-base balance was within the normal range [41]. The proportion of propionate in VFAs reached 21.64 %, significantly higher than that in Phase I. In contrast, the two experimental groups exhibited a propionate proportion of 24.51 %–23.81 %, with their VFA concentrations reaching approximately 4000 mg/L while remaining stable. Therefore, fluctuations in physicochemical parameters could not account for the performance deterioration in the control group. Conversely, severe foaming was observed in the control group during this phase. The accumulation

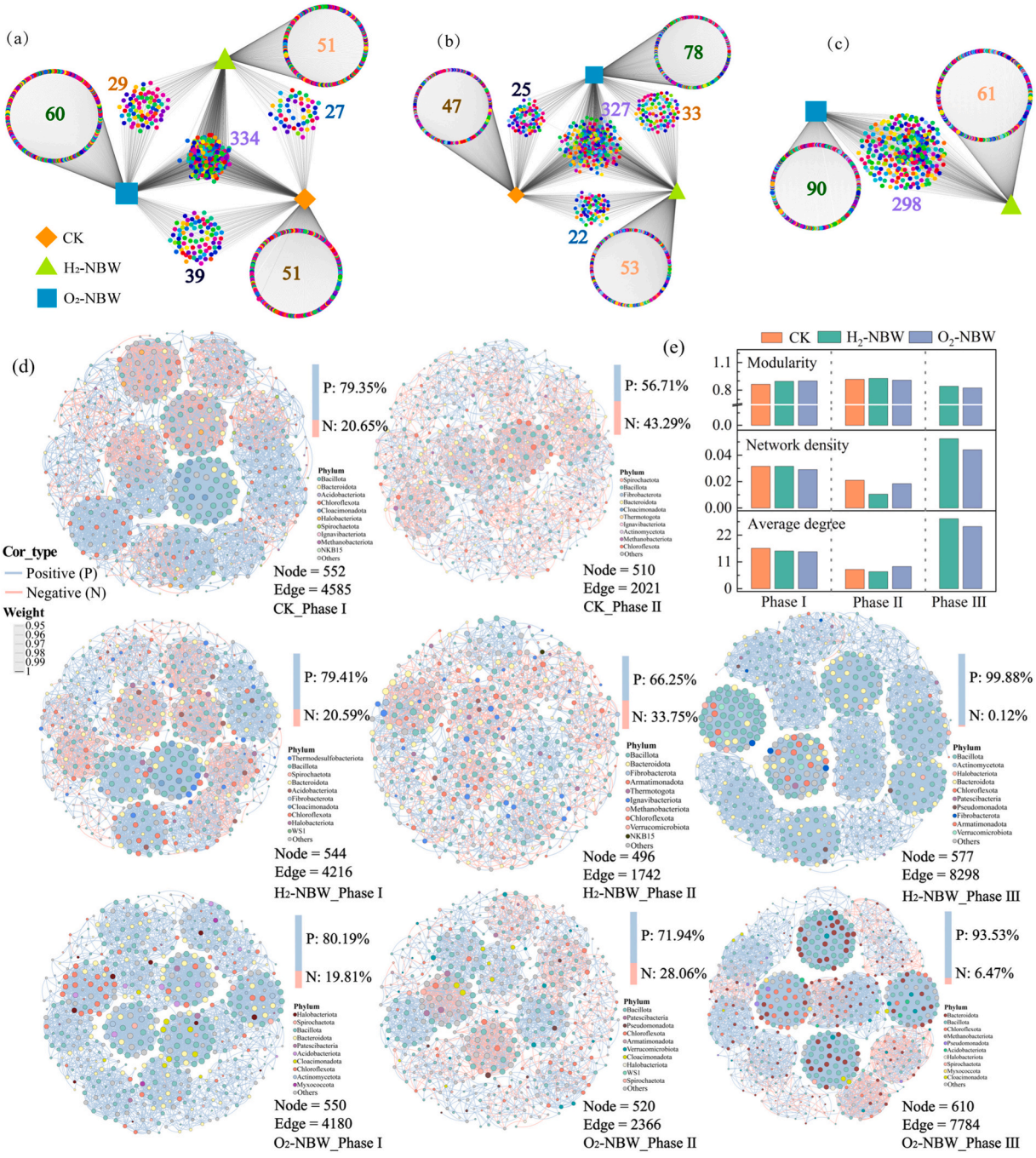


Fig. 5. Microbial community analysis: (a-c) venn diagrams of genus-level taxa across Phase I–III; (d) microbial community co-occurrence network (Spearman correlations with $|r| > 0.6$ and $p < 0.01$); and (e) topological index dynamics.

of these foams could be induced by potential foaming substances in the substrate (such as proteins, lipid derivatives, and phytohormones) or by biosurfactants accumulated within the reactor under mass transfer-limited conditions [42]. The specific causes and formation pathways require further research for clarification. However, the biogas encapsulated by liquid films cannot be smoothly discharged from the liquid phase, leading to foam formation in the reactor and at the gas-liquid interface, which exacerbates the physical isolation between substrates and microorganisms, potentially serving as a direct cause of reactor instability [43].

Given the high concentrations VFA accumulated in the two experimental groups at the end of Phase II, characterized by a persistently elevated proportion of propionate, Phase III was initiated by further increasing the OLR to validate the robustness of the NBW treatment groups. During Phase III, methane yields for the H_2-NBW and O_2-NBW groups were recorded at 157.01 ± 9.77 and 163.66 ± 8.05 mL/gVS·d, respectively, while the control group completely ceased biogas production. At this juncture, VFA concentrations in both treatment groups continued to rise, with the proportions of propionate and butyrate escalating to between 33.77 % and 51.76 %. Notably, the pH and ALK within the systems exhibited no deterioration (Fig. 4e and f). The substantial accumulation of VFAs can be attributed to enhanced hydrolysis facilitated by the addition of NBW, which established a new equilibrium between VFA generation and consumption under elevated OLR conditions. Nevertheless, nanobubbles can disrupt the hydrogen bond network of water, thereby improving water fluidity, enhancing the hydration of organic particles, and promoting substrate penetration into the inner regions of biofilms [14]. This optimization of mass transfer may elucidate why the H_2/O_2-NBW groups were able to sustain microbial metabolic activity, even amidst high VFA concentrations and elevated propionate levels [44]. In conclusion, the incorporation of NBW into reactors can lead to a sustained enhancement of the AD performance.

3.2.2. Microbial characteristics

Venn diagrams illustrate the selective enrichment patterns of functional microorganisms induced by H_2/O_2-NBW at the end of each phase. This analysis, coupled with the differential microbial results, elucidates the phenomenon of microbial community differentiation influenced by NBW. In Phase I (Fig. 5a), the H_2/O_2-NBW groups exhibited 363 common species, alongside 51 exclusive species in the H_2-NBW group and 60 in the O_2-NBW group. As the OLR increased, the number of common species diminished by 0.83 % in Phase II and by 17.91 % in Phase III for the H_2/O_2-NBW groups. Conversely, the number of exclusive species rose by 3.92 % and 30.00 % at the end of Phase II and by 19.61 % and 50.00 % at the end of Phase III, respectively (Fig. 5b and c). These succession patterns demonstrate that the addition of H_2/O_2-NBW drives microbial community restructuring, with its effects becoming increasingly pronounced at higher OLRs.

Microbial co-occurrence network analysis revealed significant structural differences in potential interaction patterns among communities across different treatment groups and operational phases (Fig. 5d–e). Specifically, the control group exhibited a decreased number of nodes, edges, proportion of positive correlations, average degree, and network density, alongside an increased modularity (from 0.90 to 0.95) by the end of Phase II compared to Phase I. This trend reflects the characteristics of sparser microbial community interactions and enhanced modularization under high OLR stress. The differences among the treatment groups were particularly significant. The H_2-NBW group showed reduced key network parameters (e.g., nodes, edges, and average degree) compared to the control group during Phases I-II (−1.45 % to −13.81 %), but exhibited higher modularity and proportion of positive correlations (+0.08 % to +16.82 %). In contrast, the network parameters (such as the number of nodes) of the O_2-NBW group were generally lower than those of the control group at the end of Phase I (−0.36 % to −8.84 %), but were higher than those of the control group at the end of Phase II (+1.96 % to +17.07 %), reflecting its unique dynamic change trajectory. By the end of Phase III, the number of nodes in the H_2-NBW group was 5.41 % lower than that in the O_2-NBW group; however, the number of connections, the proportion of positive correlations, the average degree and the network density were all higher (+1.97 % to +19.09 %), indicating that the connections between nodes in the H_2-NBW group were tighter, while the O_2-NBW group contained more nodes but had relatively loose connections. The objective differences in these network parameters clearly depict the significant structural distinctions in the potential interaction patterns of microbial communities among the control group, the H_2-NBW group, and the O_2-NBW group.

The correlations between genus-level differential microorganisms and methane yield in the H_2-NBW and O_2-NBW groups were further examined using the Mantel test. In the H_2-NBW group (Fig. 6a), methane yield was primarily positively correlated with bacterial taxa such as *DMER64*, *norank_o_Bacteroidales*, and *norank_f_Synergistaceae*, as well as archaeal taxa including *Methanobacterium*, *Methanosphaera*, *Methanolinea*, and *Candidatus_Methanomethylicus* ($p < 0.05$). In contrast, in the O_2-NBW group (Fig. 6b), methane yield showed a significant positive correlation with bacterial taxa including *unclassified_f_Oscillospiraceae*, *Sedimentibacter*, *unclassified_f_Lachnospiraceae*, and *Roseimarinus*, alongside the archaeal taxon *norank_c_Bathyarchaeia* ($p < 0.05$). Due to the inability to trace their functions, unannotated sequences (designated as 'norank' or 'unclassified') were excluded from subsequent statistical analyses and mechanistic interpretations. As illustrated in Fig. 6c, the increase in OLR from 3 gVS/(L·d) to 5 gVS/(L·d) during the long-term application of H_2-NBW resulted in a significant increase in the relative abundance of the bacterial taxon *DMER64*, with its proportion rising from 3.11 % to 14.45 % ($p < 0.05$). *DMER64* emerged as the dominant microorganism in the H_2-NBW group, playing a crucial role as a major H_2 carrier that significantly enhances interspecies H_2 transfer [45]. Among the archaeal taxa (Fig. 6d), only *Methanobacterium* exhibited a relative abundance of ≥1 %. The relative abundance of this archaeon, a typical hydrogenotrophic methanogen, increased with the rising OLR from 1.09 % to 1.41 % ($p < 0.05$) [46]. In contrast, in the O_2-NBW group, the relative abundance of *Sedimentibacter* increased with increasing OLR, from 0.98 % (Phase I) to 3.48 % (Phase III, $p < 0.05$). This bacterium is regarded as a potential key hydrolytic and acidogenic bacterium for the hydrolysis and acidogenesis of complex polymers in the AD process [47–49]. In summary, H_2-NBW enhances methane synthesis efficiency by strengthening the metabolism of key functional microorganisms involved in hydrogenotrophic methanogenesis, while O_2-NBW optimizes substrate decomposition pathways by inducing the enrichment of hydrolytic and acidogenic bacteria. Both treatments achieve increased methane yield through the directional regulation of functional microbial community succession.

3.3. PICRUSt2 function prediction

Functional predictions of relevant metabolic pathways were conducted using PICRUSt2 and subsequently annotated with the KEGG database. It is important to emphasize that these analytical results are inferences derived from 16S rRNA gene sequencing, and their conclusions require validation in the future through techniques such as metatranscriptomics or gene-specific quantification.

The results indicated that amino acid metabolism, carbohydrate metabolism, and energy metabolism were the predominant secondary metabolic pathways (Fig. 7a). Cell motility, an indicator of microbial metabolic strength, was notably high in the control, H_2-NBW, and O_2-NBW groups, with prediction abundances of 1.11 %, 1.45 %, and 1.43 %, respectively. The tertiary metabolic functions are illustrated in Fig. 7b. The order of prediction abundance for methane metabolism was O_2-NBW > H_2-NBW > control, which corresponds with the observed methane yields (Fig. 4). Pyruvate metabolism, along with starch and sucrose metabolism, is intricately linked to the hydrolysis process, with

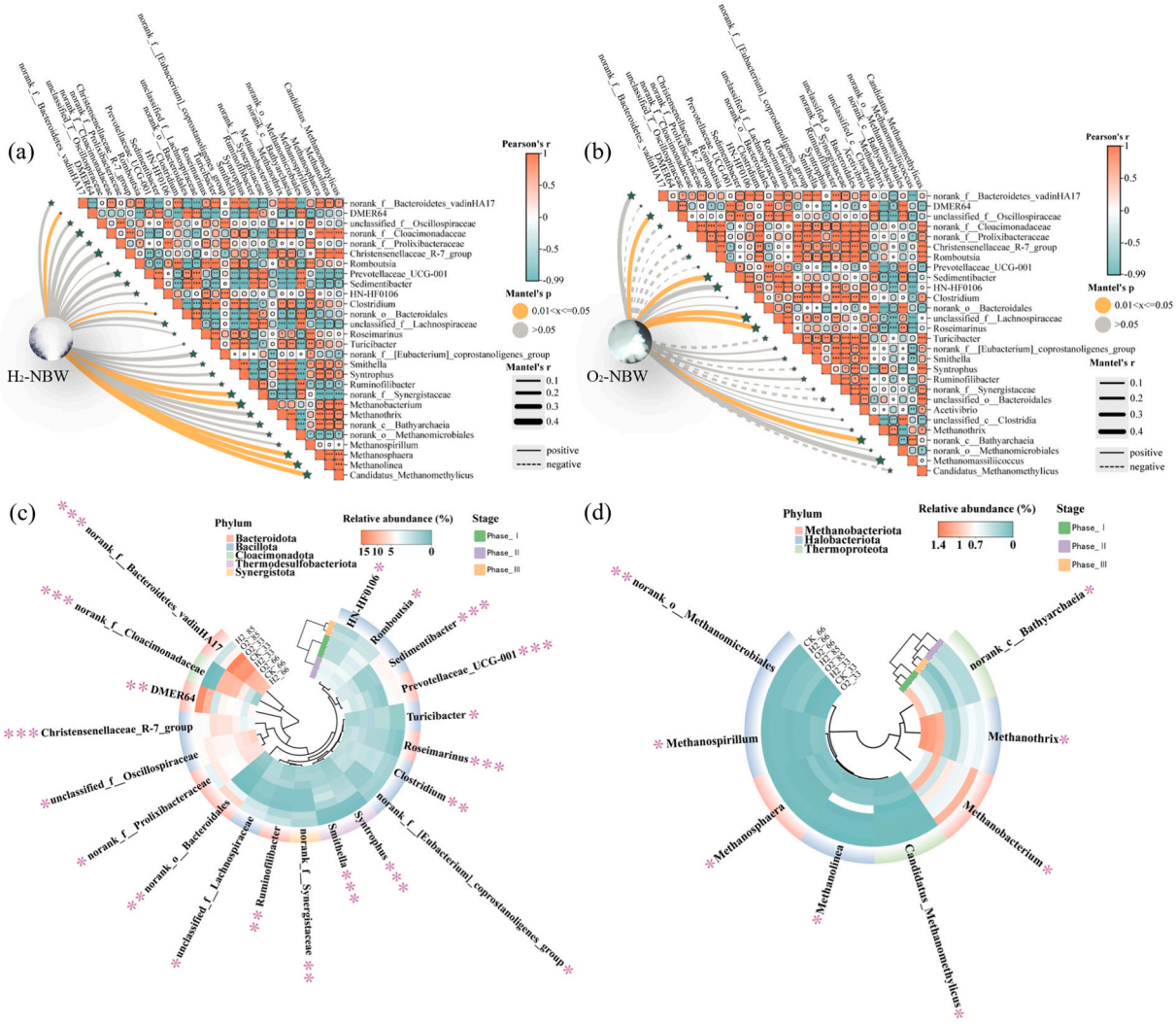


Fig. 6. Mantel test analysis linking differential microorganisms to methane yield in (a) H_2-NBW and (b) O_2-NBW groups. The heatmap employs a color gradient to indicate the strength of the Pearson correlation. The width of the connecting lines represents the Mantel's *r* values, with the color of the lines denoting the significance of *p*-values derived from permutation tests. Solid lines indicate positive correlations, while dashed lines represent negative correlations. Differential (c) bacterial and (d) archaeal taxa at genus level. Significance levels: $*p < 0.05$, $** p < 0.01$, and $***p \leq 0.001$.

their prediction abundances serving as indicators of hydrolysis intensity [50]. Compared to the control group, the prediction abundances of pyruvate metabolism and starch and sucrose metabolism in the H_2-NBW and O_2-NBW groups increased by 1.00 %–3.76 % and 0.53 %–10.28 %, respectively. Notably, the total prediction abundance of the aforementioned genes in the O_2-NBW group was elevated by 6.05 % relative to the H_2-NBW group, indicating that O_2-NBW may enhance the genetic capacity of hydrolysis-related genes, as evidenced by the increased hydrolase activity (Fig. 2) and the rise in the relative abundance of hydrolytic bacteria. Additionally, the abundance of ABC transporters is frequently utilized as a proxy for assessing the intensity of hydrolytic activity within the AD system [51]. The prediction abundance of ABC transporters in the O_2-NBW group increased by 5.64 % compared to the H_2-NBW group, further corroborating the notion that O_2-NBW may exert targeted regulation on the functional capacity of substrate hydrolysis. The citrate cycle, also referred to as the tricarboxylic acid (TCA) cycle, represents the final metabolic pathway for fats, proteins, and carbohydrates, facilitating the conversion of nutrients into energy and supplying essential intermediates necessary for maintaining cellular functions [52]. The prediction abundances of various amino acid metabolic pathways and the TCA cycle were significantly elevated in the H_2/O_2-NBW groups compared to the control group, suggesting that the introduction of NBW may have enhanced the functional potential of microbial communities within these metabolic pathways, thereby potentially supporting increased metabolic activity.

It has been reported that cellulose is hydrolyzed into glucose monomers through the action of endoglucanase and glucosidase. In contrast, hemicellulose is converted into xylo-oligosaccharides by endo-1,4-beta-xylanase via the random cleavage of the xylan backbone, which are subsequently hydrolyzed into xylose monomers [10]. Glucose is then converted to pyruvate via the Embden-Meyerhof pathway, accompanied by the production of NADH and ATP [53]. Proteins derived from cattle manure are hydrolyzed into amino acids by extracellular metalloprotease (EC 3.4.24.X) and are subsequently deaminated by transaminase (EC 2.6.1.X) to yield α-keto acids, which can further be converted to pyruvate [50]. The oxidative decarboxylation of pyruvate generates acetyl-CoA, which is ultimately converted into acetate or butyrate through the shunt metabolism of acetate kinase or butyrate kinase. As illustrated in Fig. 7c, the prediction abundance of these key enzymes in the O_2-NBW group increased by 0.96 % to 26.08 % and by 0.88 % to 18.66 % compared to the control and H_2-NBW groups, respectively. This indicates that this treatment effectively optimized substrate bioaccessibility by specifically activating the potential functional capacity of the hydrolysis and acidogenesis phases. Interestingly,

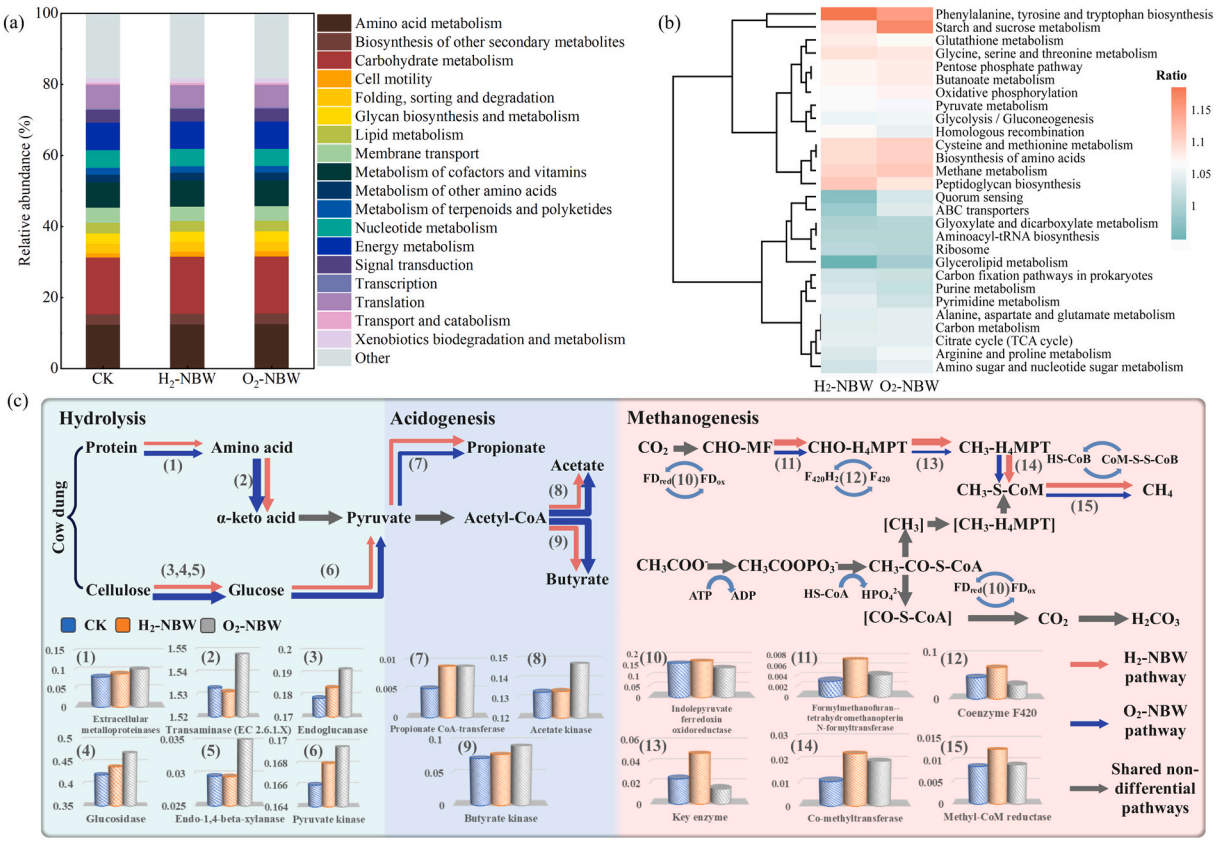


Fig. 7. Predicted microbial metabolic functions: (a) primary and secondary metabolic pathways; (b) heatmap of tertiary functional profiles (values represent the abundance ratios of H_2/O_2-NBW groups in comparison to the control); (c) dynamics of key enzyme-encoding genes involved in hydrolysis, acidogenesis, and methanogenesis, where key enzymes in the methanogenesis pathway include EC 3.5.4.27, EC 1.5.98.1, and EC 1.5.98.2.

this finding contradicts the observations from the batch experiment, which revealed that enzyme activity was higher in the H_2-NBW group than in the O_2-NBW group (Fig. 2e). This discrepancy may be attributed to fundamental differences in the experimental systems; batch systems reflect enzymatic status at the endpoint of substrate depletion, whereas semi-continuous systems reflect enzyme activity under steady-state conditions. Furthermore, O_2-NBW indirectly enhances acetogenesis by accelerating hydrolysis and acidogenesis, demonstrating greater sustainability in continuously operated systems. In contrast, H_2-NBW may stimulate acetate cleavage through CO_2 sequestration effects, leading to transient increases in acetokinase activity at the batch endpoints. This observation underscores the dynamic complexity of microbial regulation mediated by NBW.

The core reaction catalyzed by indolepyruvate ferredoxin oxidoreductase (IOR) generates acetyl-CoA while concurrently reducing ferredoxin (Fd_{red}). As a critical electron carrier, the dynamic equilibrium of Fd_{red} directly regulates metabolic pathways reliant on electron transfer. In the H_2-NBW group, the prediction abundance of IOR increased by 6.67 % compared to the control group, potentially linked to dissolved hydrogen acting as an electron donor that enhances the regeneration of Fd_{red}, thus facilitating electron transfer in indole metabolism. Conversely, in the O_2-NBW group, IOR prediction abundance decreased by 13.33 %, likely due to dissolved oxygen consuming Fd_{red}, thereby disrupting anaerobic electron transfer chains. Formylmethanofuran-tetrahydromethanopterin N-formyltransferase (EC 2.3.1.101), methenyltetrahydromethanopterin cyclohydrolase (EC 3.5.4.27), methylenetetrahydromethanopterin dehydrogenase (EC 1.5.98.1), 5,10-methylenetetrahydromethanopterin reductase (EC 1.5.98.2), co-methyltransferase (EC 2.1.1.245), and coenzyme F_{420} are integral components of hydrogenotrophic methanogenesis. The prediction abundance of these key functional genes in the H_2-NBW group increased by 7.60 %–46.63 % and by 12.09 %–119.76 % compared to the control and O_2-NBW groups, respectively (Fig. 7c), indicating that the introduction of H_2-NBW may enhance the genetic capacity for hydrogenotrophic methanogenesis. Furthermore, methyl-CoM reductase is widely recognized as a marker for methanogenesis [35]. The prediction abundance of methyl-CoM reductase (MCR) in the H_2-NBW group increased by 44.34 % and 38.10 % compared to the control and O_2-NBW groups, respectively. This suggests that H_2-NBW may enrich hydrogenotrophic methanogens harboring the MCR gene (e.g., *Methanobacterium*) and enhance the genetic capacity for methanogenesis. Such changes may improve the terminal methanogenic stage of AD by activating hydrogenotrophic methanogenesis. Conversely, the prediction abundance of MCR in the O_2-NBW group showed no significant change compared to the control group, further confirming that the micro-oxic environment induced by O_2-NBW may suppress the functional potential of methanogens, potentially due to hydroxyl radical-mediated DNA damage.

In summary, H_2/O_2-NBW optimizes AD processes through distinct metabolic regulation strategies. The addition of H_2-NBW appears to activate the genetic encoding potential of key genes involved in hydrogenotrophic methanogenesis, while the introduction of O_2-NBW emphasizes the enhancement of the hydrolysis and acidogenesis stages. Its micro-oxic properties may potentially suppress the functional capacity of methanogens. It is important to acknowledge that functional predictions are inherently less accurate than quantitative gene expression analyses. Therefore, future studies should employ hydrogen isotope tracing techniques to directly quantify H_2 conversion pathways and conduct systematic hydrogen balance accounting. Additionally, metatranscriptomic or targeted gene quantification approaches should be utilized to analyze the expression levels of key functional genes, such as *mcrA*, *fwd*, and *hyaB*. This integrated methodology will establish

definitive causal relationships at both the metabolic flux and molecular levels, thereby transcending inferences based solely on enzymatic activity and functional potential, and accurately elucidating the core mechanisms through which H_2/O_2-NBW regulates AD.

4. Conclusion

This study elucidates the potential and mechanisms underlying the targeted enhancement of AD using H_2/O_2-NBW. Batch experiment results demonstrated that H_2-NBW primarily activated dehydrogenase, coenzyme F_{420}, and acetokinase, leading to an enhancement in methane yield of 4.21 %–16.98 % compared to the control group. In contrast, O_2-NBW facilitated a positive feedback loop of substrate supply and metabolic conversion through hydroxyl radical-mediated deconstruction of cellulose crystallinity, which decreased by 39.81 %, and enhanced hydrolytic enzyme activity, resulting in methane yield and cellulose removal efficiency increases of 3.37 %–13.43 % and 2.75 %–132.13 %, respectively, compared to the control group. PLS-PM and RFA results indicated that the primary influencing factors for methane production in the H_2-NBW and O_2-NBW groups were coenzyme F_{420} and cellulase, respectively. Furthermore, 100 %H_2-NBW and 80 %O_2-NBW were identified as optimal additive dosages. Semi-continuous experiments revealed that the H_2/O_2-NBW groups maintained a methane yield of 157.01–163.66 mL/gVS·d at a high OLR of 5 gVS/(L·d). The microbial co-occurrence network analysis indicated that the addition of NBW significantly altered microbial interaction patterns. Among them, the H_2-NBW group enriched functional bacteria related to hydrogenotrophic methanogenesis and acetogenesis, including *DMER64* and *Methanobacterium*, while O_2-NBW targeted the regulation of *Sedimentibacter*, *which* belong to hydrolytic and acidogenic bacteria. Functional predictions indicated that H_2/O_2-NBW enhanced key metabolic pathways related to hydrogenotrophic methanogenesis and hydrolysis/acidogenesis, respectively. Future research should focus on comparing the effects of direct gas addition with those of NBW, as well as quantitatively analyzing the direct contribution of dissolved hydrogen in H_2-NBW to methane production through the CO_2 reduction pathway. Such investigations will significantly enhance our understanding of the potential mechanisms underlying the effects of nanobubble water.

CRediT authorship contribution statement

Linpei Han: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Hengbing Zhou:** Visualization, Methodology, Data curation. **Lei Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Wenjie Ye:** Visualization, Conceptualization. **Peng Xu:** Visualization, Conceptualization. **Yongli Liu:** Visualization, Methodology. **Bin Zhan:** Visualization, Methodology. **Xuya Peng:** Visualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.171515.

Data availability

The authors do not have permission to share data.

References

[1] National Bureau of Statistics of China, China Statistical Yearbook. https://www.stats.gov.cn/sj/ndsj/2024/indexch.htm, 2024. (Accessed 22 July 2025).
[2] H. Lin, H. Jiao, H. Lin, X. Xu, The evolution of policies for the resource utilization of livestock manure in China, Agriculture 15 (2025) 153, https://doi.org/10.3390/agriculture15020153.
[3] J. Zou, X. Liu, S. Xu, M. Chen, Q. Yu, J. Xie, Combined hydrothermal pretreatment of agricultural and forestry wastes to enhance anaerobic digestion for methane production, Chem. Eng. J. 486 (2024) 150313, https://doi.org/10.1016/j.cej.2024.150313.
[4] J. Huang, Y. Zhu, T. Liu, S. Sun, J. Ren, A. Wu, H. Li, A novel wet-mechanochemical pretreatment for the efficient enzymatic saccharification of lignocelluloses: small dosage dilute alkali assisted ball milling, Energy Convers. Manag. 194 (2019) 46–54, https://doi.org/10.1016/j.enconman.2019.04.078.
[5] K. Paritosh, M. Yadav, N. Kesharwani, N. Pareek, O. Parthiba Karthyikeyan, V. Balan, V. Vivekanand, Strategies to improve solid state anaerobic bioconversion of lignocellulosic biomass: an overview, Bioresour. Technol. 331 (2021) 125036, https://doi.org/10.1016/j.biortech.2021.125036.
[6] S. Wei, The application of biotechnology on the enhancing of biogas production from lignocellulosic waste, Appl. Microbiol. Biotechnol. 100 (2016) 9821–9836, https://doi.org/10.1007/s00253-016-7926-5.
[7] H. Baniamerian, P. Ghofrani-Isfahani, P. Tsapekos, M. Alvarado-Morales, M. Shahrokhi, I. Angelidaki, Multicomponent nanoparticles as means to improve anaerobic digestion performance, Chemosphere 283 (2021) 131277, https://doi.org/10.1016/j.chemosphere.2021.131277.
[8] T. Hou, H. Song, Z. Cui, C. He, L. Liu, P. Li, G. Li, Q. Zhang, Z. Zhang, Z. Lei, Y. V. Litti, Y. Jiao, Nanobubble technology to enhance energy recovery from anaerobic digestion of organic solid wastes: potential mechanisms and recent advancements, Sci. Total Environ. 931 (2024) 172885, https://doi.org/10.1016/j.scitotenv.2024.172885.
[9] C. He, H. Song, L. Liu, P. Li, M. Kumar Awasthi, G. Xu, Q. Zhang, Y. Jiao, C. Chang, Y. Yang, Enhancement of methane production by anaerobic digestion of corn straw with hydrogen-nanobubble water, Bioresour. Technol. 344 (2022) 126220, https://doi.org/10.1016/j.biortech.2021.126220.
[10] X. Wang, T. Yuan, Z. Guo, H. Han, Z. Lei, K. Shimizu, Z. Zhang, D.-J. Lee, Enhanced hydrolysis and acidification of cellulose at high loading for methane production via anaerobic digestion supplemented with high mobility nanobubble water, Bioresour. Technol. 297 (2020) 122499, https://doi.org/10.1016/j.biortech.2019.122499.
[11] Y. Fan, X. Yang, Z. Lei, Z. Zhang, M. Kobayashi, Y. Adachi, K. Shimizu, Alleviation of ammonia inhibition via nano-bubble water supplementation during anaerobic digestion of ammonia-rich swine manure: buffering capacity promotion and methane production enhancement, Bioresour. Technol. 333 (2021) 125131, https://doi.org/10.1016/j.biortech.2021.125131.
[12] M. Alheshibri, J. Qian, M. Jehannin, V.S.J. Craig, A history of nanobubbles, Langmuir 32 (2016) 11086–11100, https://doi.org/10.1021/acs.langmuir.6b02489.
[13] S. Zhou, K.R. Marcelino, S. Wongkiew, L. Sun, W. Guo, S.K. Khanal, H. Lu, Untapped potential: applying microbubble and nanobubble technology in water and wastewater treatment and ecological restoration, ACS EST Eng. 2 (2022) 1558–1573, https://doi.org/10.1021/acsestengg.2c00117.
[14] W. Chuenchart, R. Karki, T. Shitanaka, K.R. Marcelino, H. Lu, S.K. Khanal, Nanobubble technology in anaerobic digestion: a review, Bioresour. Technol. 329 (2021) 124916, https://doi.org/10.1016/j.biortech.2021.124916.
[15] J. Zhao, Y. Li, R. Dong, Recent progress towards in-situ biogas upgrading technologies, Sci. Total Environ. 800 (2021) 149667, https://doi.org/10.1016/j.scitotenv.2021.149667.
[16] Y. Fan, Z. Lei, Z. Guo, W. Huang, D. Wang, X. Wang, Z. Zhang, K. Shimizu, Enhanced solubilization of solid organics and methane production by anaerobic digestion of swine manure under nano-bubble water addition, Bioresour. Technol. 299 (2020) 122512, https://doi.org/10.1016/j.biortech.2019.122512.
[17] X. Yang, J. Nie, D. Wang, Z. Zhao, M. Kobayashi, Y. Adachi, K. Shimizu, Z. Lei, Z. Zhang, Enhanced hydrolysis of waste activated sludge for methane production via anaerobic digestion under N2-nanobubble water addition, Sci. Total Environ. 693 (2019) 133524, https://doi.org/10.1016/j.scitotenv.2019.07.330.
[18] E. Mancini, H. Tian, I. Angelidaki, I.A. Fotidis, The implications of using organic-rich industrial wastewater as biomethanation feedstocks, Renew. Sust. Energ. Rev. 144 (2021) 110987, https://doi.org/10.1016/j.rser.2021.110987.
[19] N. Joy, G. Senthilkumar, P.G. Logesh, R. Hariharan, Investigation on physical properties of nanobubbled water using gas water circulation method: environmental perspective, Environ. Qual. Manag. 32 (2023) 161–165, https://doi.org/10.1002/tqem.21956.
[20] Y. Fan, Z. Lei, X. Yang, M. Kobayashi, Y. Adachi, Z. Zhang, K. Shimizu, Effect of nano-bubble water on high solid anaerobic digestion of pig manure: focus on digestion stability, methanogenesis performance and related mechanisms, Bioresour. Technol. 315 (2020) 123793, https://doi.org/10.1016/j.biortech.2020.123793.

[21] T. Lyu, F. Wu, N. Nasar, X. Li, P. Jarvis, Y. Bajón-Fernández, Exploring nanobubble technology for enhanced anaerobic digestion of thermal-hydrolysis pre-treated sewage sludge, Bioresour. Technol. Rep. 27 (2024) 101939, https://doi.org/10.1016/j.biteb.2024.101939.
[22] L. Wang, J. Ali, Z. Wang, N.A. Oladoja, R. Cheng, C. Zhang, G. Mailhot, G. Pan, Oxygen nanobubbles enhanced photodegradation of oxytetracycline under visible light: synergistic effect and mechanism, Chem. Eng. J. 388 (2020) 124227, https://doi.org/10.1016/j.cej.2020.124227.
[23] J. Park, R.D.A. Cayetano, Y. Kwon, G.-B. Kim, Y. Jo, S.-H. Kim, Improved sludge anaerobic digestion capacity by dynamic membrane and alkaline-thermal pretreatment: long-term continuous operation and techno-economic analysis, Chem. Eng. J. 474 (2023) 145735, https://doi.org/10.1016/j.cej.2023.145735.
[24] Y. Kwon, J. Park, G.-B. Kim, Y. Jo, S. Park, S.-H. Kim, Anaerobic digestion of sewage sludge using anaerobic dynamic membrane bioreactor under various sludge composition and organic loading rates, Bioresour. Technol. 384 (2023) 129275, https://doi.org/10.1016/j.biortech.2023.129275.
[25] Y. Peng, L. Li, P. Yang, H. Liu, W. Ye, Z. Xue, X. Peng, X. Wang, Integrated genome-centric metagenomic and metaproteomic analyses unravel the responses of the microbial community to ammonia stress, Water Res. 242 (2023) 120239, https://doi.org/10.1016/j.watres.2023.120239.
[26] W. Ye, L. Li, Z. Tang, Y. Peng, P. Yang, X. Peng, F. Zhen, Biochar and modified magnetic biochar enhanced anaerobic digestion of swine wastewater under ammonia stress: performance and microbial dynamics, J. Environ. Chem. Eng. 12 (2024) 111969, https://doi.org/10.1016/j.jece.2024.111969.
[27] G.S. Rana, N. Naudiyal, R. Bal, A.K. Verma, A. Dubey, B.R. Singh, T.P. Singh, A comprehensive extraction-precipitation method and characterization study to optimize biogas slurry enzyme recovery, Waste Biomass Valoriz. (2025), https://doi.org/10.1007/s12649-025-02884-2.
[28] Z. Wang, X. Gui, H. Luan, Z. Li, Y. Zeng, Y. Zheng, Permanganate pretreatment improves the production of short chain fatty acids from waste activated sludge at pH10: performance and mechanism, Sep. Purif. Technol. 362 (2025) 131692, https://doi.org/10.1016/j.seppur.2025.131692.
[29] X. Wang, T. Yuan, Z. Lei, M. Kobayashi, Y. Adachi, K. Shimizu, D.-J. Lee, Z. Zhang, Supplementation of O2-containing gas nanobubble water to enhance methane production from anaerobic digestion of cellulose, Chem. Eng. J. 398 (2020) 125652, https://doi.org/10.1016/j.cej.2020.125652.
[30] G. Wu, L. Zou, F. Huang, B. Wang, S. Huang, X. Shen, S. Chen, J. Zhu, Effect of humic substances derived from pastoral areas in Zoige plateau on photodegradation of sulfamethoxazole and ciprofloxacin, Process. Saf. Environ. Prot. 159 (2022) 819–829, https://doi.org/10.1016/j.psep.2022.01.049.
[31] P. Yang, Y. Peng, H. Liu, D. Wu, R. Yuan, X. Wang, L. Li, X. Peng, Multi-scale analysis of the foaming mechanism in anaerobic digestion of food waste: from physicochemical parameter, microbial community to metabolite response, Water Res. 218 (2022) 118482, https://doi.org/10.1016/j.watres.2022.118482.
[32] Z. Jin, S. Lv, L. Zhang, S. Liu, W. Fan, Nanobubble and slow-release hydrogen delivery methods for interfacial remediation of copper-induced toxicity in aquatic environments, Langmuir 41 (2025) 17590–17601, https://doi.org/10.1021/acs.langmuir.5c01213.
[33] S. Fu, S. Lian, I. Angelidaki, R. Guo, Micro-aeration: an attractive strategy to facilitate anaerobic digestion, Trends Biotechnol. 41 (2023) 714–726, https://doi.org/10.1016/j.tibtech.2022.09.008.
[34] A. Graham, C. Thorn, M. McDonagh, C. O'Donnell, S. Nolan, S.F. Kirwan, S. O'Connor, C.O. Nzeteu, A.C.V. Montoya, A. Bartle, A. Hall, C. Abberton, R. Friel, S.M. Waters, V. O'Flaherty, Development and in-vitro assessment of novel oxygen-releasing feed additives to reduce enteric ruminant methane emissions, Sci. Total Environ. 963 (2025) 177598, https://doi.org/10.1016/j.scitotenv.2024.177598.
[35] P.S. Adam, G.E. Kolyfetis, T.L.V. Bornemann, C.E. Vorgias, A.J. Probst, Genomic remnants of ancestral methanogenesis and hydrogenotrophy in Archaea drive anaerobic carbon cycling, Sci. Adv. 8 (2022), https://doi.org/10.1126/sciadv.abm9651.
[36] M. Zhou, Y. Wang, Y. Wang, T. Tu, J. Zhang, X. Wang, G. Zhang, H. Huang, B. Yao, H. Luo, X. Qin, Sequential pretreatment with hydroxyl radical and manganese peroxidase for the efficient enzymatic saccharification of corn stover, Biotechnol. Biofuels 17 (2024) 136, https://doi.org/10.1186/s13068-024-02583-5.
[37] S. Yaparatne, J. Morón-López, D. Bouchard, S. Garcia-Segura, O.G. Apul, Nanobubble applications in aquaculture industry for improving harvest yield, wastewater treatment, and disease control, Sci. Total Environ. 931 (2024) 172687, https://doi.org/10.1016/j.scitotenv.2024.172687.
[38] Z. Wang, S. Wang, Y. Hu, B. Du, J. Meng, G. Wu, H. Liu, X. Zhan, Distinguishing responses of acetoclastic and hydrogenotrophic methanogens to ammonia stress in mesophilic mixed cultures, Water Res. 224 (2022) 119029, https://doi.org/10.1016/j.watres.2022.119029.
[39] Q. Qu, L. Cheng, P. Wang, C. Fang, H. Li, J. Ding, H. Wan, G. Guan, Guanidine-functionalized basic binuclear poly(ionic liquid)s for low partial pressure CO2 fixation into cyclic carbonate, Sep. Purif. Technol. 339 (2024) 126682, https://doi.org/10.1016/j.seppur.2024.126682.
[40] W. Tian, L. Gu, S. Li, L. Zhu, W. Li, R. Deng, F. Liu, Facilitating digester recovery from acid inhibition at high organic load rates by limited calcium peroxide addition, ACS Sustain. Chem. Eng. 10 (2022) 8184–8195, https://doi.org/10.1021/acssuschemeng.2c01780.
[41] D. Wu, M. Zhang, S. Hu, Y. Liu, F. Zhen, H. Liu, Y. Sun, Multidimensional insights into efficiency-stability trade-off in anaerobic digestion from organic overloading stress: kinetic behavior, energy harvesting, and genome-centric metagenomics microbial self-adaptation, Water Res. 284 (2025) 123990, https://doi.org/10.1016/j.watres.2025.123990.
[42] W. Ye, L. Li, P. Yang, Z. Zhang, L. Han, R. Yuan, X. Peng, Mechanism of extracellular polymeric substance production and its contribution to anaerobic digestion foaming: insights from cellular signalling, Chem. Eng. J. 500 (2024) 157377, https://doi.org/10.1016/j.cej.2024.157377.
[43] P. Yang, L. Li, Y. Zhou, S. Zhang, Z. Zhang, Y. Peng, H. Liu, W. Ye, X. Peng, Insights into the effect of extracellular polymeric substances on anaerobic digestion foaming: from perspectives of composition, hydrophobicity, and functional groups, Chem. Eng. J. 476 (2023) 146812, https://doi.org/10.1016/j.cej.2023.146812.
[44] X. Hao, R. Liu, M.C.M. van Loosdrecht, D. Cao, Batch influences of exogenous hydrogen on both acidogenesis and methanogenesis of excess sludge, Chem. Eng. J. 317 (2017) 544–550, https://doi.org/10.1016/j.cej.2017.02.093.
[45] X. Meng, Q. Cao, Y. Sun, S. Huang, X. Liu, D. Li, 16S rRNA genes- and metagenome-based confirmation of syntrophic butyrate-oxidizing methanogenesis enriched in high butyrate loading, Bioresour. Technol. 345 (2022) 126483, https://doi.org/10.1016/j.biortech.2021.126483.
[46] X. Li, Y. Wang, Y. He, X. Wang, H. Guo, T. Zhu, P. Sun, B.-J. Ni, Y. Liu, Ultrasound-enhanced peracetic acid oxidation improves short-chain fatty acids production from waste activated sludge, Chem. Eng. J. 478 (2023) 147214, https://doi.org/10.1016/j.cej.2023.147214.
[47] P. Hu, M. Xiao, N. Wang, S. Zhang, J. Shi, J. Shi, T. Tang, L. Liu, Metagenome reveals the possible mechanism that microbial strains promote methanogenesis during anaerobic digestion of food waste, Environ. Res. 251 (2024) 118723, https://doi.org/10.1016/j.envres.2024.118723.
[48] J. Ma, J. Pan, Y. Zhang, Z. Yao, J. Yu, J. Luo, R. Shen, M.K. Awasthi, L. Zhao, Alleviating "inhibited steady-state" in anaerobic digestion of poultry manure by bentonite amendment: performance evaluation and microbial mechanism, Bioresour. Technol. 360 (2022) 127519, https://doi.org/10.1016/j.biortech.2022.127519.
[49] Y. Zhu, T. Lyu, D. Li, Z. Zhang, J. Guo, X. Li, W. Xiong, R. Dong, S. Wang, Process mechanisms of nanobubble technology enhanced hydrolytic acidification of anaerobic digestion of lignocellulosic biomass, Chem. Eng. J. 480 (2024) 147956, https://doi.org/10.1016/j.cej.2023.147956.
[50] X. Chen, Y. Xie, C. Zang, W. Wang, J. Tian, B. Wen, Z. Cui, X. Yuan, Biological pretreatment and co-digestion strategies enhance the methane yield of waste wool, Chem. Eng. J. 502 (2024) 158089, https://doi.org/10.1016/j.cej.2024.158089.
[51] X. Chen, Z. Cui, Y. Zhao, N. Zhu, Y. Liu, Z. Hu, X. Yuan, Synergistic mechanism of substrate hydrolysis and methanogenesis under "gradient anaerobic digestion" process, Energy Convers. Manag. 309 (2024) 118443, https://doi.org/10.1016/j.enconman.2024.118443.
[52] Z.-L. Gao, Y.-W. Cui, Y. Sui, H.-J. Yang, Unveiling the mechanism of static magnetic field effects on carbon flux into polyhydroxyalkanoate synthesis by Haloferax mediterranei, Chem. Eng. J. 476 (2023) 146647, https://doi.org/10.1016/j.cej.2023.146647.
[53] D. Ngoc Pham, A. Duc Nguyen, D. Hoang Anh Mai, E. Yeol Lee, Development of a novel methanotrophic platform to produce ectoine from methane and lignocellulose-derived sugars, Chem. Eng. J. 463 (2023) 142361, https://doi.org/10.1016/j.cej.2023.142361.